Statistical models and BMP

Sasha D. Hafner

April 2024

Overview

Purpose is to check error distribution for BMP data.

Analysis

Load data

```
library(data.table)
```

```
BMP <- fread('../data/BMP.csv')
```

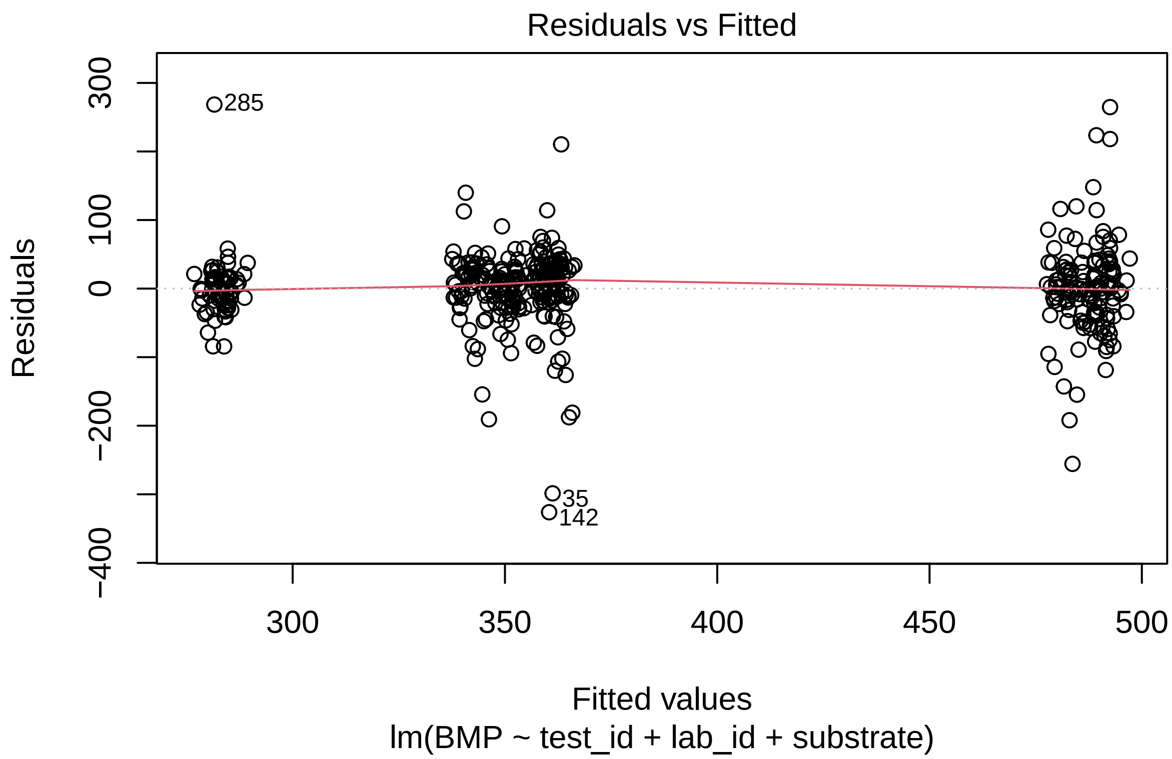
Fit linear fixed-effects model to make diagnostic plots easier.

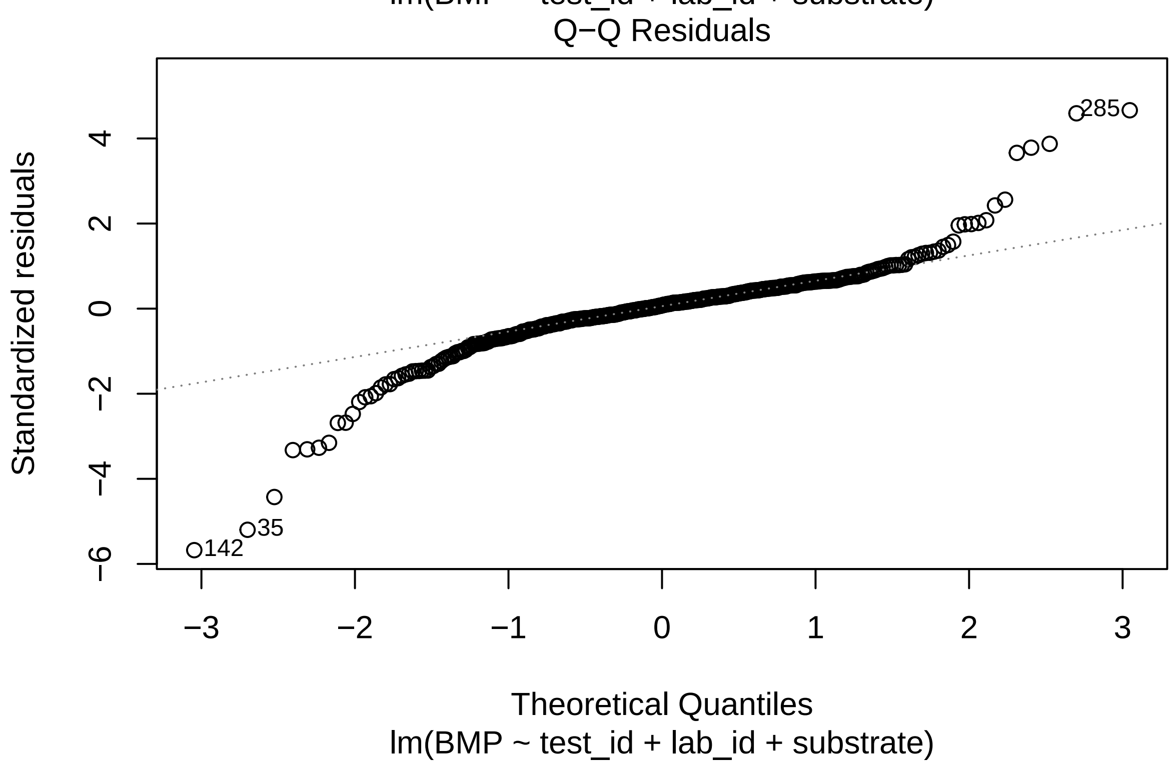
```
mod1 <- lm(BMP ~ test_id + lab_id + substrate, data = BMP)
summary(mod1)
```

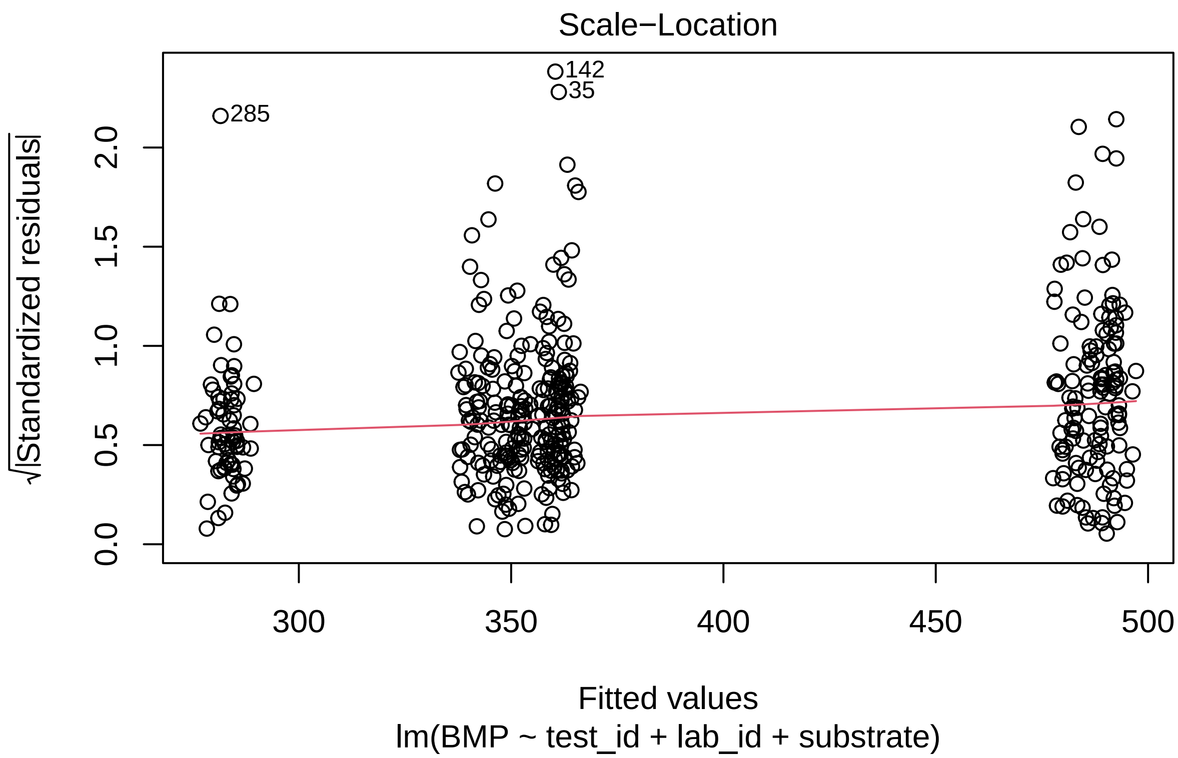
```
##
## Call:
## lm(formula = BMP ~ test_id + lab_id + substrate, data = BMP)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -326.18  -19.82    4.40   26.44  268.44
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 328.38231   13.94219  23.553  < 2e-16 ***
## test_id       0.07454    0.06160   1.210   0.2270
## lab_id        0.21266    0.25401   0.837   0.4029
## substrateSA  18.86068   10.17120   1.854   0.0644 .
## substrateSB  19.63269   10.39526   1.889   0.0596 .
## substrateSC 140.05748    7.27720  19.246  < 2e-16 ***
## substrateSD -67.68683    9.27534  -7.298 1.45e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 58.03 on 425 degrees of freedom
## Multiple R-squared:  0.6198, Adjusted R-squared:  0.6145
## F-statistic: 115.5 on 6 and 425 DF,  p-value: < 2.2e-16
```

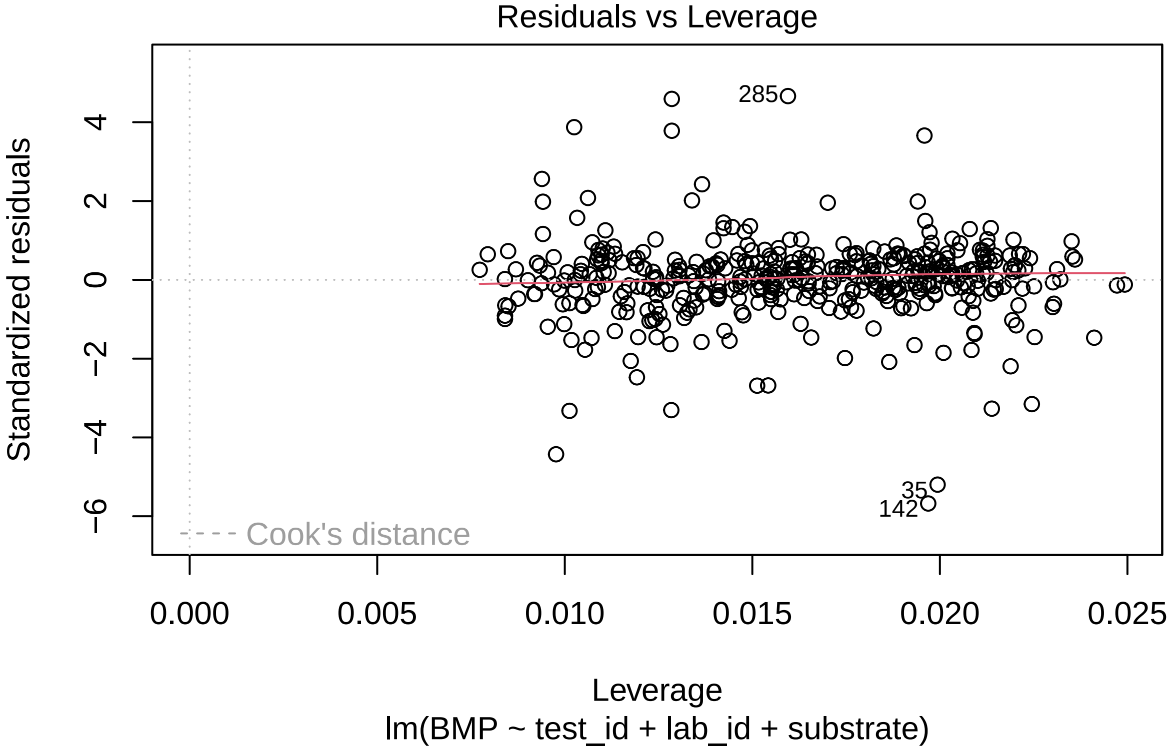
Diagnostics

```r
plot(mod1, ask = FALSE)
```









Try lognormal.

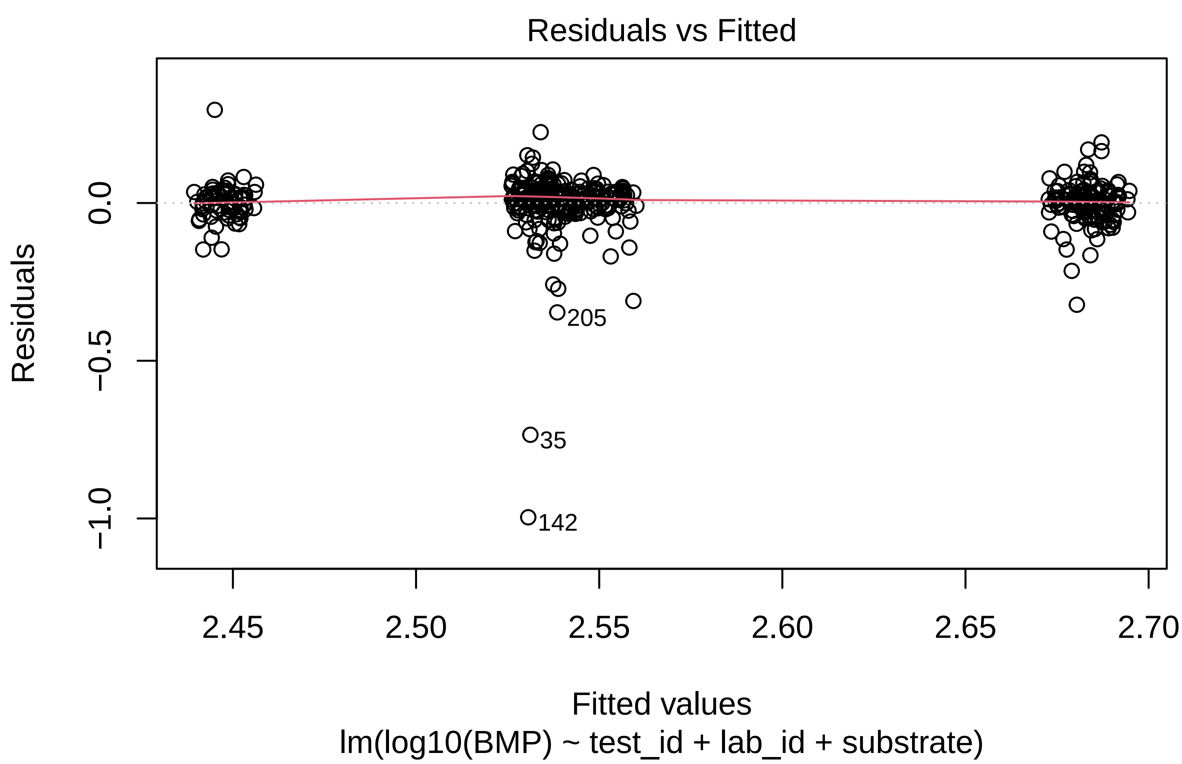
```
mod2 <- lm(log10(BMP) ~ test_id + lab_id + substrate, data = BMP)
summary(mod2)
```

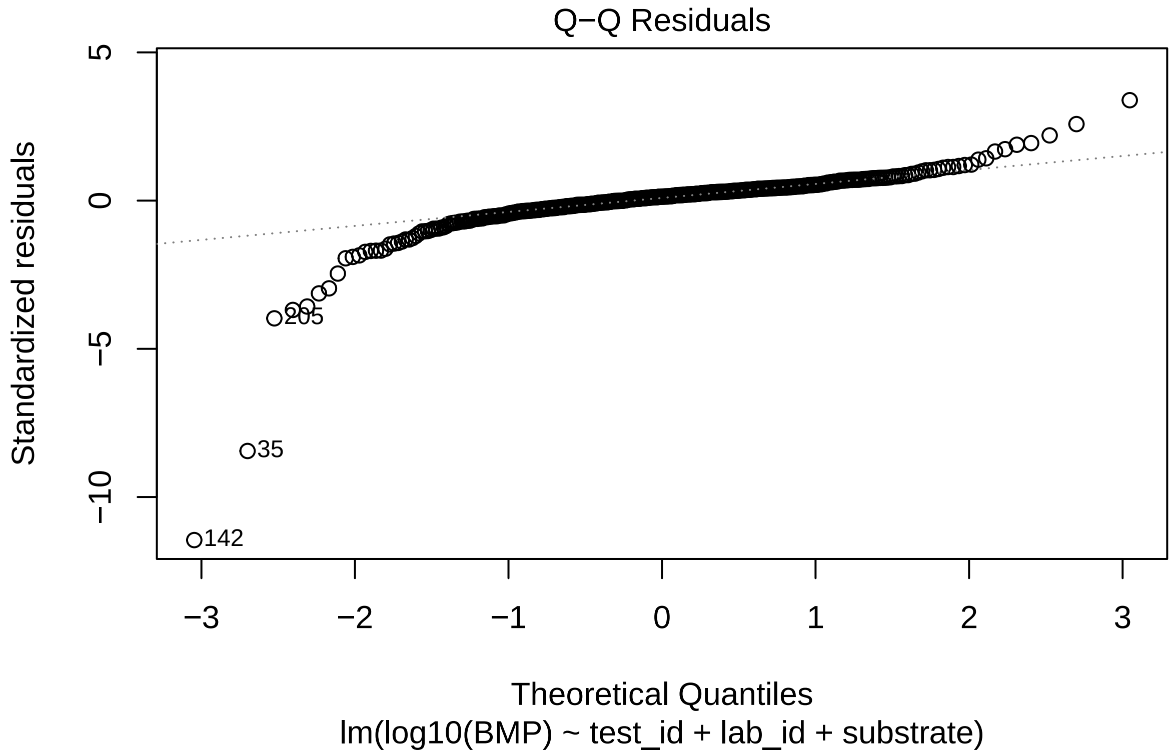
```
##
```

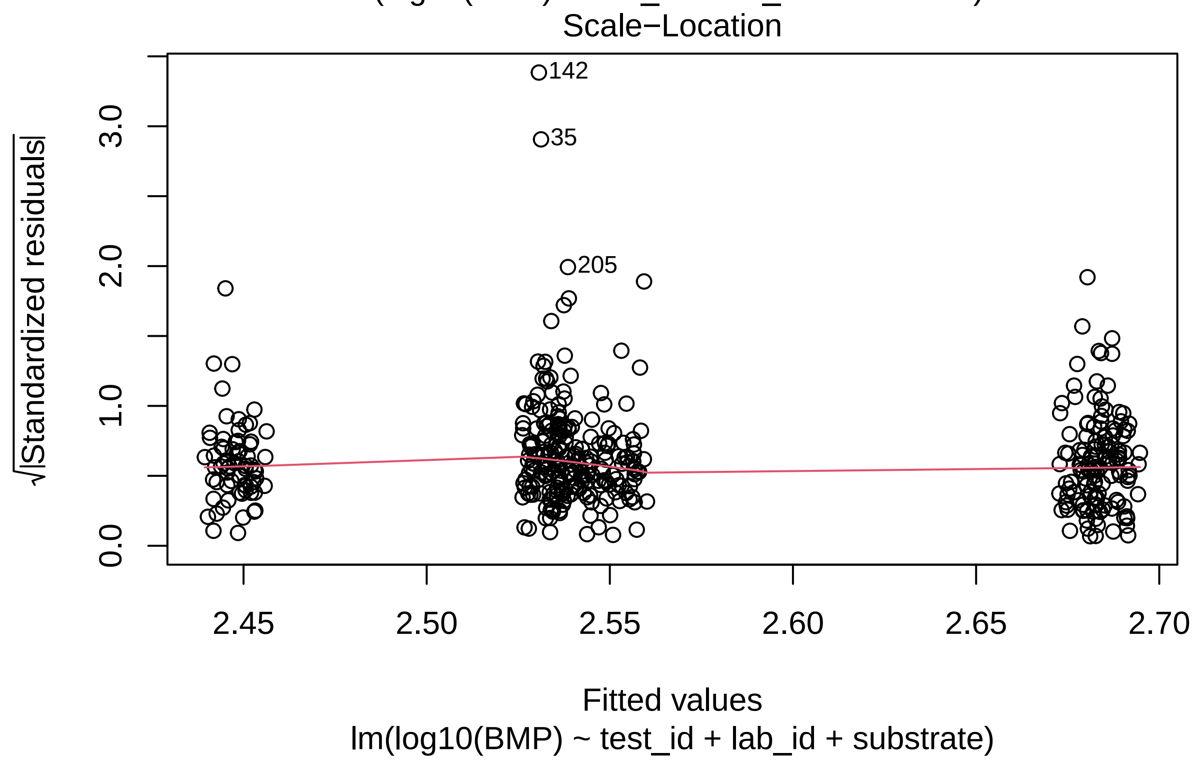
```
## Call:
## lm(formula = log10(BMP) ~ test_id + lab_id + substrate, data = BMP)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.99623 -0.01987  0.01174  0.03555  0.29530
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.519e+00  2.111e-02 119.326  < 2e-16 ***
## test_id      5.288e-05  9.327e-05   0.567    0.571
## lab_id       3.775e-04  3.846e-04   0.981    0.327
## substrateSA  2.078e-02  1.540e-02   1.350    0.178
## substrateSB  2.503e-04  1.574e-02   0.016    0.987
## substrateSC  1.466e-01  1.102e-02  13.309  < 2e-16 ***
## substrateSD -9.179e-02  1.404e-02  -6.536 1.81e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08787 on 425 degrees of freedom
## Multiple R-squared:  0.4731, Adjusted R-squared:  0.4656
## F-statistic: 63.59 on 6 and 425 DF,  p-value: < 2.2e-16
```

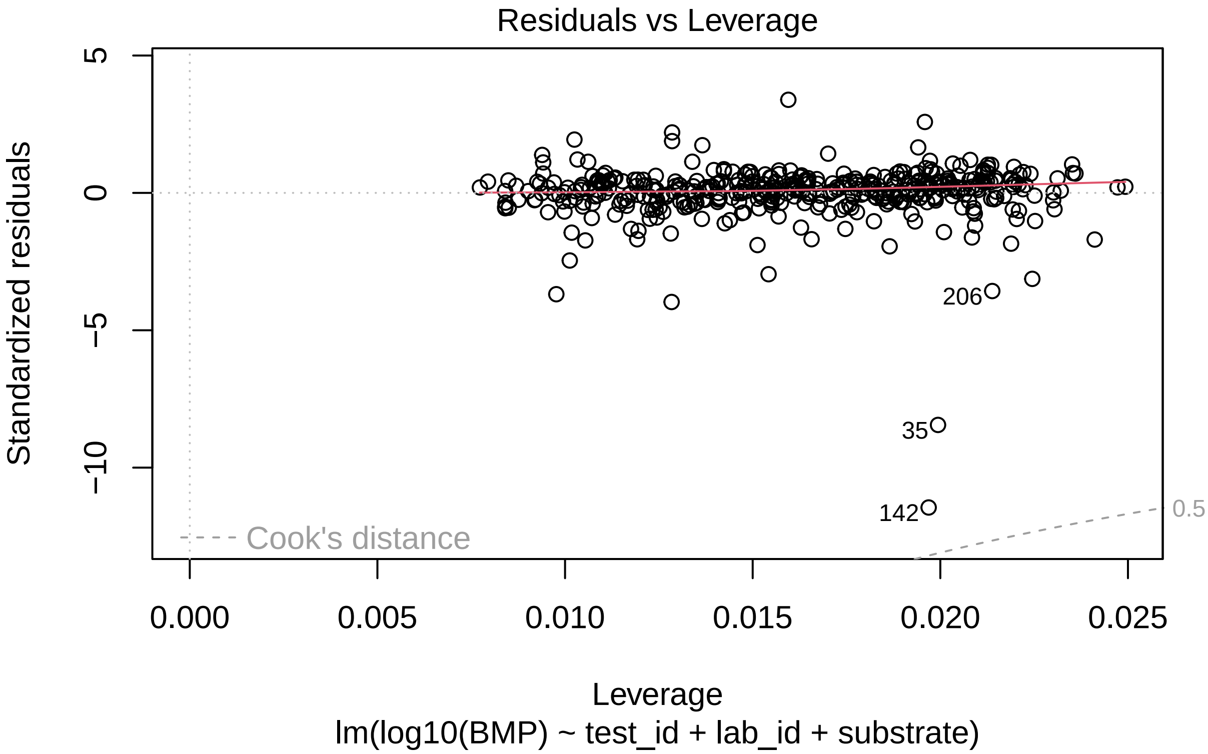
Diagnostics

```
plot(mod2, ask = FALSE)
```




Q−Q Residuals
Standardized residuals
5
0
−5
−10
205
35
142
−3
−2
−1
0
1
2
3
Theoretical Quantiles
lm(log10(BMP) ~ test_id + lab_id + substrate)


Scale−Location
√|Standardized residuals|
0.0
1.0
2.0
3.0
142
35
205
2.45
2.50
2.55
2.60
2.65
2.70
Fitted values
lm(log10(BMP) ~ test_id + lab_id + substrate)



Conclusion

No major evidence of heteroscedastiscity. But some deviation from normality in errors. Log transformation does not help issue that much. But it is most likely related to unusual extreme values.